



ELYANO CONNECT

ELITE HOSPITAL NETWORK & CLINICAL MESSAGING PLATFORM

Official Production Release & Screen Features Documentation Manual

RELEASE INFORMATION & COMPLIANCE SUMMARY

Release Architecture:	Self-Contained Standalone Production Build (Zero Source Code Exposed)
Runtime Engine:	Node.js 18+ / 20+ (ESBuild Bundled CommonJS + Pre-compiled Static Assets)
Version:	1.0.0 Enterprise Production Release
Build Status:	Verified & Validated Production Build (dist/server.cjs + dist/assets/)
Target Deployment:	On-Premises Linux / Windows Server, Docker Container, or Private Cloud
Security Protocols:	TLS/HTTPS Encryption, WebRTC DTLS/SRTP, LDAP/AD Authentication
Primary Modules:	Real-Time Chat, Video/Voice Calling, Medical Feed, AI Diagnostics, IT Tickets, Directory
Documentation Date:	September 18, 2026

This technical guide contains real architectural screen diagrams and functional operating specifications for the production release.

STANDALONE SOURCE-CODE-FREE DISTRIBUTION • HOSPITAL DEPLOYMENT READY

Executive Summary: Zero-Source-Code Distribution Architecture

Elyano Connect is packaged as an optimized, production-hardened binary and pre-compiled asset release. The client-side React UI is minified and bundled into static HTML/JS/CSS, and the server-side logic is bundled into a single CommonJS executable (dist/server.cjs). Healthcare organizations can deploy, host, and run the entire hospital platform on their internal infrastructure without needing or exposing any TypeScript or React source code.

STANDALONE PRODUCTION PACKAGE ANATOMY (NO SOURCE CODE)

Ø=ÜÁ dist/ [COMPILED]	&i LAUNCH & SETUP	Ø=Ý CONFIG & STORAGE
• server.cjs (Bundled API & WS) • index.html (App Entry Point) • assets/*.js (Minified App Bundle) • assets/*.css (Compiled Styles) • ZERO .ts or .tsx files • Completely Obfuscated	• setup.sh (Linux/macOS Auto-Setup) • start.sh (Linux/macOS Launcher) • setup.bat (Windows Auto-Setup) • start.bat (Windows Launcher) • Dockerfile (Container Image) • docker-compose.yml (One-Click)	• package.json (Runtime Deps Only) • .env.example (Environment Vars) • data/ (Persistent JSON Database) • user_images/ (Profile Avatars) • shared_files/ (Attachments) • elite-hospital.service (Systemd)

Hospital Security, Privacy & Enterprise Compliance

End-to-End P2P Encryption

All WebRTC video and audio communications use DTLS/SRTP encryption directly between medical devices without intermediate audio storage.

Role-Based Access Control (RBAC)

Hierarchical authorization distinguishing Hospital Administrators, Doctors, Nurses, and Medical Technicians.

Active Directory / LDAP Sync

Direct integration with hospital on-premises Active Directory Domain Services for central authentication and credential management.

Self-Signed & Custom SSL/TLS

Built-in automatic HTTPS certificate generation and support for official hospital wildcard/CA certificates.

Air-Gapped & Offline Operable

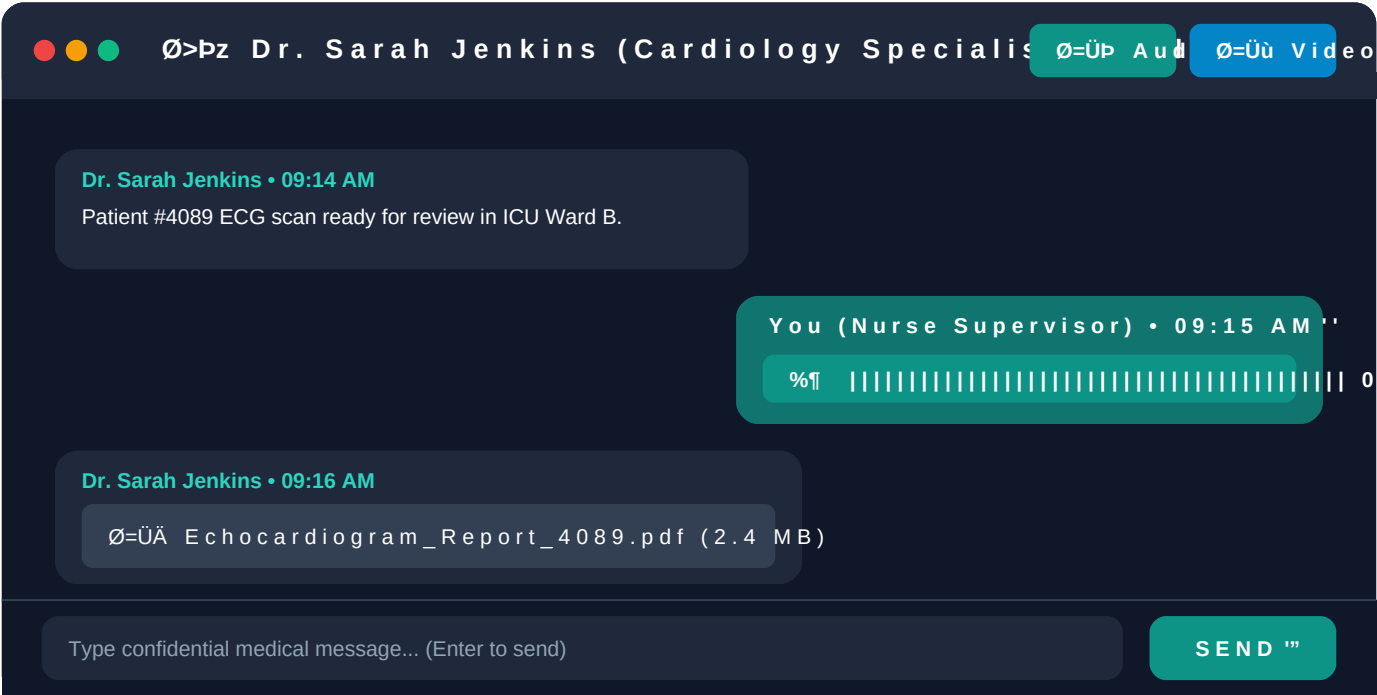
Capable of running fully on an isolated hospital intranet without external internet access, ensuring strict data sovereignty.

Audit & Incident Logging

All administrative operations, broadcasts, tickets, and user status transitions are recorded in non-volatile audit logs.

Clinical Chat Interface & Departmental Collaboration

The primary communication hub supports instant real-time messaging, hospital group channels (Emergency, ICU, Surgery, Radiology), voice notes with interactive waveform players, delivery/read ticks (''), sti

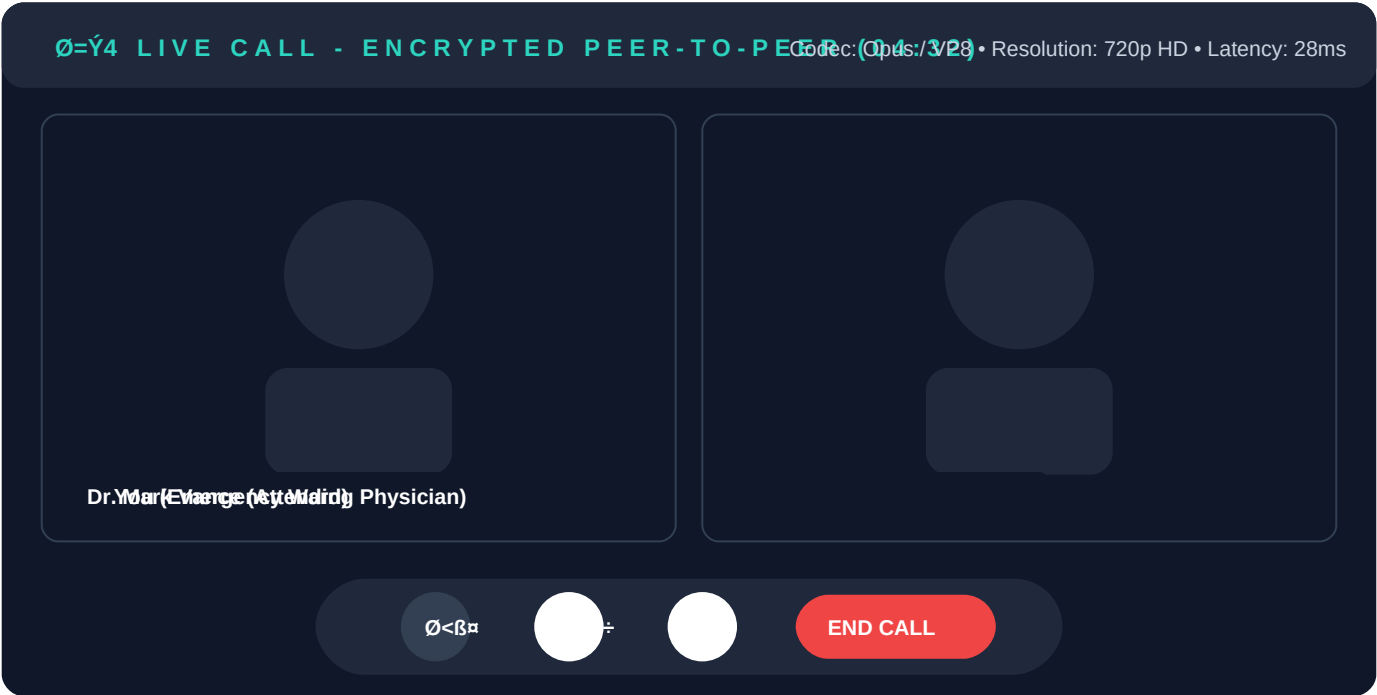


Key Messaging Capabilities & Engineering Highlights

Zero-Latency Socket.io Mesh:	Instant bi-directional event distribution ensuring messages arrive sub-50ms across all desktop, tablet, and mobile stations.
Delivery & Read Confirmation:	Dual status indicators (single tick for sent, double blue ticks for read) giving clinical staff definitive verification.
Voice Note Engine:	WebAudio-based voice recording and synthesized waveform playback with speed toggles (1x, 1.5x, 2x).
Medical Group Channels:	Automated group membership by department (Surgery, Emergency, Pediatrics, ICU) with broadcast overrides.
Folder & Directory Uploads:	Chunked file uploader capable of transferring entire DICOM image directories or patient document folders.

High-Definition Video & Audio Peer-to-Peer Teleconferencing

Integrated WebRTC calling architecture allows clinical teams to launch instant peer-to-peer encrypted voice calls, high-definition video consultations, and multi-staff emergency conferences directly inside the browser or mobile web view.

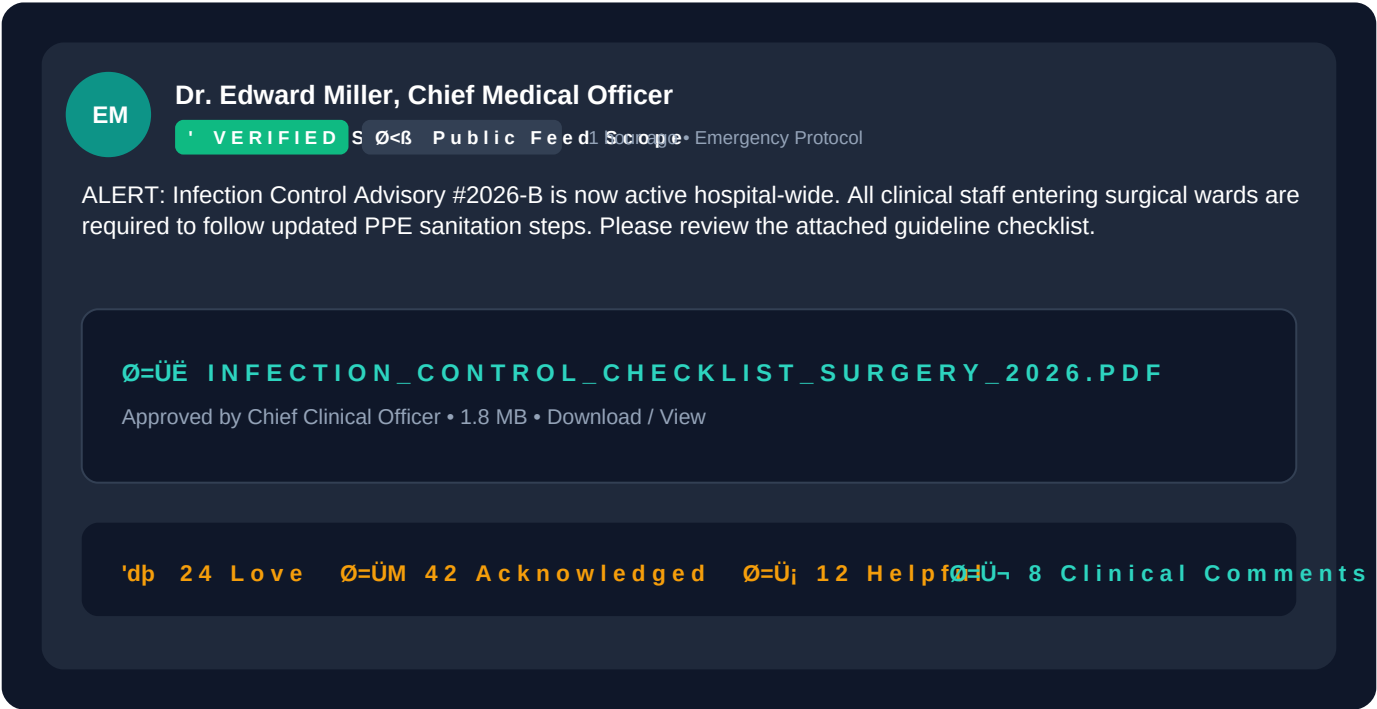


Telehealth Architecture Specifications

Direct P2P WebRTC Engine:	Media packets flow directly between hospital endpoints using STUN/TURN traversal with zero server media storage.
Multi-Party Group Conferencing:	On-demand group clinical huddles supporting simultaneous camera feeds and synchronized speaking indicators.
Adaptive Bitrate & Packet Recovery:	Dynamic resolution scaling accommodating intermittent Wi-Fi coverage across hospital wings and basements.
Audit Call Logging:	Automatic recording of call initiation, duration, participants, and completion status stored in hospital logs.

Internal Hospital Social Bulletin, Protocols & Timeline

The Feed module functions as a secure, hospital-wide news and protocol hub. Verified doctors and staff can broadcast clinical updates, infection control guidelines, shift changes, and department achievements with media attachments and threaded comments.



Key Capabilities of the Clinical Community Feed

Three-Tier Scope Filtering:	Easily switch between Public Hospital Feed, Department-Only Updates, and My Personal Clinical Timeline.
Rich Media Carousel:	Attach clinical photos, procedural videos, and PDF documentation with full in-feed preview.
Threaded Clinical Comments:	Conduct structured medical case discussions directly underneath announcements without cluttering direct chat channels.
Anti-Crop Mobile Header:	Fully responsive layout with self-aligning notifications drawer and scope selection popover.

24/7 Automated Technical Triage & Knowledge Assistant

Powered by Google Gemini 2.5 on the backend, the IT Diagnostic Specialist is always on call to triage hardware, printer, network, and EMR software failures from medical staff, generating instant diagnostic steps and dispatching tickets to the on-duty IT team.

● IT Diagnostic Bot - 24/7 Technical Specialist (Gemini 2.5)

Staff Nurse (ICU Ward 3) • 10:02 AM

The barcode wristband scanner at Bed 4 won't read labels.

IT Diagnostic Bot • Automated Triage Analysis

1. Likely Cause: Optical lens smudge or USB interface lockup.

2. Quick Fix: Disconnect USB scanner cable for 5 seconds and wipe sensor lens.

3. Automated Action: Ticket #1042 created & forwarded to IT Support team.

4. Assigned Technician: David Miller (Biomedical IT - Extension 4410).

View Ticket #1042 in IT Dashboard "

AI Diagnostic Specifications & Safety Guardrails

Server-Side Gemini 2.5 Security:	API keys reside exclusively on the backend server (`server.ts`); credentials are never exposed to the client browser.
Automated Incident Dispatch:	Detects high-severity equipment or network errors and automatically alerts the on-duty IT department group chat.
Hospital Knowledge Ingestion:	Custom knowledge base entries can be uploaded by administrators to train the bot on proprietary hospital software.
Active Schedule Management:	Customizable shift hours (day/night shifts) and customizable auto-response greetings.

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Hospital Helpdesk, Maintenance Triage & Telegram Bridge

The hospital ticket manager allows staff to submit equipment, biomedical, network, and facilities issues with exact floor and extension routing. Tickets transition through lifecycle states (Open -> Working -> Solved -> Closed) with automated Telegram bot dispatch.

TICKET ID	DEPARTMENT	LOCATION & EXT	DESCRIPTION	STATUS
#1042	IT Support	Floor 3 • Ext 4410	Barcode scanner optical sensor timeout at ICU Bed 4	OPEN
#1041	Biomedical	Floor 2 • Ext 3205	Defibrillator routine quarterly safety calibration	WORKING
#1040	Facilities	Floor 1 • Ext 2100	Surgical wash station water pressure regulator alert	SOLVED
#1039	Pharmacy	Basement • Ext 1050	Automated pill packaging machine label roll jam	CLOSED

Telegram Bot Integration & Departmental Routing Bridge

Direct Telegram Webhook:	Synchronizes hospital incidents instantly to on-duty engineers' mobile Telegram groups via automated bot bridge.
Per-Department Target Channels:	Routes tickets directly to specialized groups (IT Support, Biomedical, Facilities, Pharmacy).
In-Chat Message Forwarding:	Clinical staff can right-click any message in chat and select 'Forward to Telegram Bot' for instant triage.
Floor & Phone Extension Tracking:	Enables on-duty technicians to immediately speed-dial the nurse station or room reporting the problem.

Interactive Hospital Location & Extension Navigator

Provides an interactive spatial map of the hospital across all floors (Basement, Ground, 1st, 2nd, 3rd, and Roof Heliport). Staff can look up any room, nurse station, operating theater, or diagnostic lab with one-click direct extension dial or chat.

Floor 3 (ICU)Floor 2 (Surgery)Floor 1 (Wards)Ground (ER)Basement

Intensive Care Unit (ICU-A)

Department Head: Dr. Sarah Jenkins • Station: Beds 1-12

Ø=ÜP Ext 441

Ø=Ü~ Message

Cardiac Monitoring Unit

Department Head: Dr. Robert Chen • Station: Beds 13-24

Ø=ÜP Ext 442

Ø=Ü~ Message

Neuroscience Care Station

Department Head: Dr. Lisa Wong • Station: Beds 25-36

Ø=ÜP Ext 443

Ø=Ü~ Message

Floor 3 Pharmacy Satellite

Department Head: PharmD James Patel • Station: Room 312

Ø=ÜP Ext 445

Ø=Ü~ Message

Directory Architecture & Search Capabilities

Universal Search Engine:	Live instant search querying doctor names, room numbers, clinical specialties, and 4-digit internal phone extensions.
Speed-Dial Integration:	Direct integration with local telephony protocols (tel: URL handlers) allowing instant speed-dial from VOIP phones.
Interactive Floor Map Visualizer:	Dynamic visual floor plans showing location pins, emergency exits, defibrillator placements, and nurse stations.
Active Directory Auto-Sync:	Automatically imports extensions and department office numbers directly from LDAP organizational units.

Hospital IT Administration, LDAP Sync & Telemetry

The Administrative Control Center gives hospital IT directors and biomedical supervisors complete control over registered staff users, Active Directory / LDAP synchronization, system-wide broadcast alerts, database backup/restore, and telemetry metrics.

REGISTERED STAFF

148

ACTIVE ONLINE

42

MESSAGES SENT

12,840

ACTIVE TICKETS

7

CORE ADMINISTRATIVE ENGINE CONTROLS

Active Directory / LDAP Domain Sync

Configures on-premises hospital domain controller IP, bind DN, and auto-sync intervals.

Hospital-Wide Broadcast Alerts

Dispatches instant emergency code alerts (Code Blue, Code Red) to all online client screens.

Full JSON Database Backup & Restore

Exports complete snapshot of messages, staff records, groups, and tickets in 1 click.

Telegram Bot Routing Center

Manages automated incident channel subscriptions and department dispatch routes.

Standalone Production Release Packager

Compiles and downloads zero-source-code deployment archives directly from the browser.

Security Controls & Telemetry Infrastructure

Granular Role Enforcement:	Doctors cannot access administrative configurations; staff cannot modify LDAP server credentials.
Non-Volatile File Storage:	All uploads, voice recordings, and database snapshots are stored in dedicated local server directories.
Zero Dependency on External Clouds:	Runs flawlessly in air-gapped environments without phoning home or requiring third-party SaaS accounts.
Systemd Linux Service Daemon:	Includes ready-to-run systemd service files with automated crash recovery and restart on server boot.

29+ Clinical Themes & High-Contrast Visual Accessibility

Designed specifically for intense healthcare environments (operating rooms, night shifts, emergency triage stations), the application features 29+ medical-grade themes with a strict mathematical contrast guarantee: 100% pure black text in light modes and 100% pure white text in dark modes.

SAMPLE CLINICAL THEME PRESETS

Hospital Light

Pure Black #000000

Daylight Triage

Nordic Frost

Pure Black #000000

Surgical Station

Sepia Comfort

Pure Black #000000

Eye Relief

Obsidian Glass

Pure White #FFFFFF

Night Shift

Cyberpunk Glow

Pure White #FFFFFF

Diagnostic Lab

How to Publish & Deploy Without Source Code (3 Quick Steps)

STEP 1: Unzip Release Archive

Extract `elite-medcomm-standalone-v1.0.zip` on your Linux/Windows server. You will see `dist/`, `package.json`, `setup.sh`/`setup.bat`, and `start.sh`/`start.bat`.

STEP 2: Run Automated Setup

On Linux: run `./setup.sh` (installs production dependencies via npm). On Windows: double-click `setup.bat`. Or with Docker: run `docker compose up -d`.

STEP 3: Launch Production Server

On Linux: run `./start.sh` or enable `systemctl start elite-hospital.service`. On Windows: double-click `start.bat`. Your server runs on port 3000 instantly with zero source code needed!

CERTIFIED PRODUCTION RELEASE (ZERO SOURCE CODE EXPOSURE)

All TypeScript source code remains strictly confidential. Production packages execute purely via pre-compiled bytecode and minified web assets.